

2

(a)

Ans//

$$C = \begin{bmatrix} 0.25 & 0.20 \\ 0.20 & 0.25 \end{bmatrix}$$

EIGEN VALUES:

$$\begin{vmatrix} 0.25 - \lambda & 0.2 \\ 0.2 & 0.25 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (\lambda - 0.25)^2 - 0.2^2 = 0$$

$$\Rightarrow (\lambda - 0.25 + 0.2)(\lambda - 0.25 - 0.2) = 0$$

$$\Rightarrow \lambda_1 = 0.45 \quad \lambda_2 = 0.05$$

EIGEN VECTORS:

For $\lambda_1 = 0.45$

$$\begin{bmatrix} -0.2 & 0.2 \\ 0.2 & -0.2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow v_1 = v_2$$

$$\therefore e_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

For $\lambda_2 = 0.05$

$$\begin{bmatrix} 0.2 & 0.2 \\ 0.2 & 0.2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow v_1 = -v_2$$

$$\therefore e_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

 $\therefore \lambda_1 > \lambda_2$,the first principal component is $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

$$\begin{aligned}
 \text{Percentage of variance explained by PC}_1 &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \\
 &= \frac{0.45}{0.50} \\
 &= 90\% //
 \end{aligned}$$

To pr

PROJECTION

$$G_{n \times 1} \quad x \in \mathbb{R}^{784}$$

Step (1) : Compute covariance matrix $C \in \mathbb{R}^{784 \times 784}$

Step 2 : Compute its eigen values, sort in decreasing order

$$u_1 > u_2 \dots > u_{784}$$

Step 3 : Retain only first k eigen values vectors corresponding to eigen values

$$U_k = \begin{bmatrix} | & | & & | \\ e_1 & e_2 & \dots & e_k \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{784 \times k}$$

Step 4: Center the input:

$$\bar{x} = x - \mu \quad [\mu \in \mathbb{R}^{784}]$$

Step 5: Projection,

$$Z = U_k^T (x - \mu) \quad [Z \in \mathbb{R}^k]$$